

Laptop Selection Criteria in MCDM/AHP Models

1. Introduction

In MCDM and AHP models for laptop selection, the most commonly used criteria cluster into a few recurring groups: **price/cost**, **brand image**, **technical performance**, **storage/memory**, **display/monitor**, and broader **physical or peripheral features** (Çakır & Pekkaya, 2020). The most laptop-specific paper in this corpus identifies six main criteria and reports that **brand image** ranks first, while **brand image plus price** account for about two-thirds of total priority (Çakır & Pekkaya, 2020). That same study also treats **speed**, **storage**, **monitor**, and **other/peripheral properties** as distinct criterion groups, showing the typical structure of laptop decision hierarchies rather than a single flat checklist (Çakır & Pekkaya, 2020).

Adjacent consumer-electronics studies show a very similar pattern. Smartphone AHP-TOPSIS models repeatedly use **price**, **brand**, **RAM**, **internal storage**, **technical specifications**, **user-related features**, and **physical properties**, which supports the transferability of these criterion families to notebook selection when laptop-specific evidence is thin (Goswami & Behera, 2021; Lam et al., 2023). More generally, MCDM applications in product and service selection rely on hierarchies of criteria, sub-criteria, and alternatives, with AHP most often used to derive criterion weights from pairwise comparisons (Jawad et al., 2023; Yannis et al., 2020; Villalba et al., 2024).

The literature is stronger on **which criterion families recur** than on a single universal ranking. Rankings vary by population and model design, but cost, performance, brand, and quality-related attributes dominate across domains (Hazza et al., 2022; Rasmussen et al., 2022; Balasbaneh et al., 2025).

Do MCDM or AHP laptop-selection models commonly use price, brand, and technical specification criteria?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on recurring laptop-selection criteria in MCDM models

The consensus is strong that MCDM/AHP laptop-selection models repeatedly use **price/cost**, **brand/reputation**, and **technical-specification** criteria, often expanded into storage, display, and peripheral sub-criteria (Çakır & Pekkaya, 2020; Goswami & Behera, 2021; Lam et al., 2023).

2. Methods

The search ran over more than 170 million research papers in Consensus, spanning Semantic Scholar, PubMed, and other scholarly sources. The workflow first identified a very large candidate set for laptop-selection criteria in MCDM and AHP, then expanded through citation crawling, adjacent terminology such as notebook and consumer electronics, and related device domains including smartphones, cloud services, and supplier selection.

A total of 240 papers were screened by the deep-search pipeline, 169 passed the relevance threshold, and the top 50 were included for synthesis. Priority was given to papers that explicitly enumerated criteria, described hierarchical criterion structures, or reviewed weighting practices in AHP and related MCDM methods. Because laptop-specific studies were few, adjacent product-selection and technology-selection papers were used to identify recurrent criterion families and methodological patterns.

Search Strategy

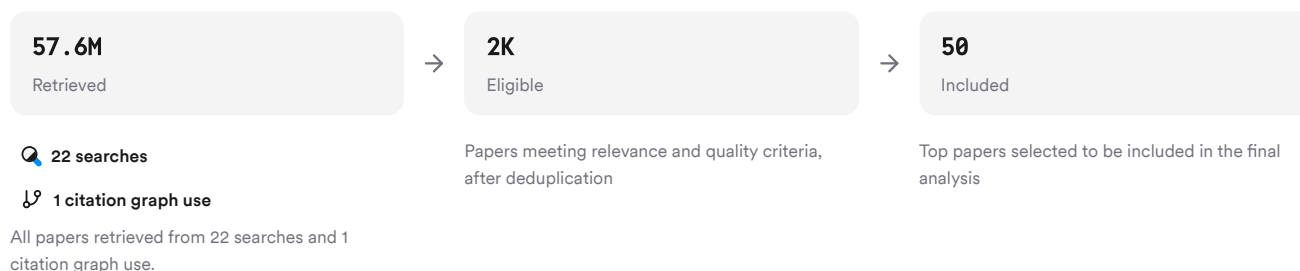


FIGURE 2 Deep-search screening flow for included MCDM papers

Searches combined laptop-specific queries with adjacent consumer-electronics, criteria-weighting, hybrid-MCDM, and critique-focused searches to recover both direct and transferable evidence.

3. Results

3.1 Key Papers

Three papers anchor this corpus for the specific question. One is directly about **laptop selection criteria and priorities**; two others are adjacent consumer-electronics studies that explicitly enumerate product-selection criteria in AHP-TOPSIS frameworks and help confirm which criterion families recur across devices (Çakır & Pekkaya, 2020; Goswami & Behera, 2021; Lam et al., 2023).

Paper	Summary
(Çakır & Pekkaya, 2020)	Laptop model with brand, price, speed, storage, monitor criteria (Çakır & Pekkaya, 2020)
(Dahooie et al., 2021)	Mobile-phone AHP-TOPSIS emphasizes technical specs and sub-criteria (Lam et al., 2023)
(Gyani et al., 2022)	Smartphone model uses price, storage, RAM, brand (Goswami & Behera, 2021)

FIGURE 3 Foundational papers for recurring device-selection criteria

3.2 Recurring Criterion Families

Across the corpus, the most reusable criterion families are **economic, brand/reputation, technical capability, quality/usability**, and **physical/design** dimensions. In the laptop-specific study, these appear as price, brand image, speed, storage, monitor, and other/peripheral properties (Çakır & Pekkaya, 2020). In smartphone studies, they appear as price, internal storage, RAM, brand, technical specification, user-related features, and physical properties (Goswami & Behera, 2021; Lam et al., 2023).

A broader product-ranking literature also emphasizes extracting and weighting **product features** rather than relying only on generic star ratings, reinforcing that feature-level criteria are central to MCDM-based selection models (Dahooie et al., 2021). Reviews across MCDM application domains likewise show that AHP is most often used to assign weights to such criteria once they have been defined (Shao et al., 2020; Villalba et al., 2024).

3.3 Typical Hierarchy Structure

AHP models usually do not treat laptop selection as a single-layer ranking problem. They decompose the problem into a **goal, criteria, sub-criteria**, and **alternatives**, which is why laptop studies often report both broad criterion groups and more granular sub-features (Jawad et al., 2023; Altubaishe & Desai, 2023).

Dimension	Typical Form in MCDM Models	Citations
Goal	Select best laptop/notebook	(Çakır & Pekkaya, 2020)
Main criteria	Price, brand, specs, display	(Çakır & Pekkaya, 2020; Goswami & Behera, 2021)
Sub-criteria	RAM, storage, processor, battery	(Lam et al., 2023; Pramanik et al., 2021)
Alternatives	Competing laptop models	(Youssef, 2020; Zakeri et al., 2023)

FIGURE 4 Common hierarchy used in AHP laptop-selection studies

This hierarchy explains why some papers report broad labels like “technical specification,” while others list concrete sub-criteria such as RAM or processor.

3.4 Ranking Patterns and Caveats

The strongest laptop-specific ranking in the corpus places **brand image** first and shows **price** as highly interactive with other criteria (Çakır & Pekkaya, 2020). Adjacent studies, however, often rank **technical specification** first, with **cost**, **battery power**, and **processor** among the top sub-criteria (Lam et al., 2023). Other smartphone models use only four major criteria—**price**, **internal storage**, **RAM**, and **brand**—showing that parsimonious models are common when alternatives are similar (Goswami & Behera, 2021).

Methodological reviews add an important caution: criteria rankings depend strongly on how weights are elicited, how many comparisons are required, and whether criteria are formally selected or mainly chosen by experts (Pamucar et al., 2018; Balasbaneh et al., 2025; Theunissen et al., 2024). Many MCDM studies still under-document the criteria-selection stage, even when weighting is carefully executed (Theunissen et al., 2024).

Results Timeline

Criterion modeling in this area has moved from classic AHP weighting toward hybrid and fuzzy variants, while keeping similar core criterion families.

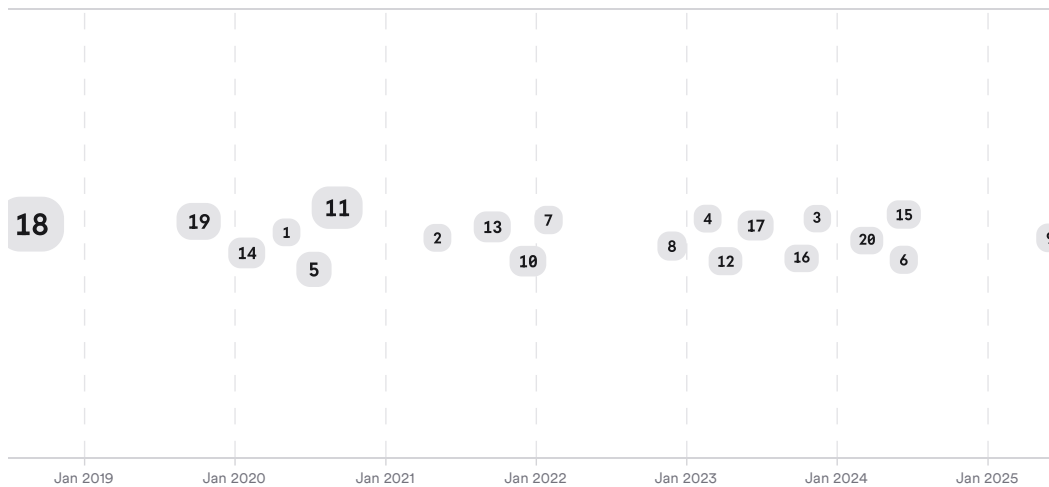


FIGURE 5 Timeline of included MCDM studies; larger markers indicate more citations

The timeline shows methodological diversification rather than a shift in what gets measured. Newer studies add fuzzy logic, hybrid weighting, or data-driven screening, but still organize decisions around cost, quality/performance, and related feature sets (Liou et al., 2021; Rahardjo et al., 2023).

Top Contributors

Type	Name	Papers
Author	D. Pamucar	[f19beaabe47855e396d12891544de9c7][9561792c142c5c1e91b244fe91029b0f]
Author	A. E. Youssef	[8d6f1b5a68885719897ecc7a5dc0e423]
Author	Imane Tronnebati	[7ce9c71461515079a3931d07145cf9eb][d8738bebf8545425ac14a6416ee34763]
Journal	<i>Sustainability</i>	[7ce9c71461515079a3931d07145cf9eb][d8738bebf8545425ac14a6416ee34763] [93634bafdded5a35a57a7effba3f04fe][9f28c914872352f4ab0217ddb15b3d0f]
Journal	<i>IEEE Access</i>	[8d6f1b5a68885719897ecc7a5dc0e423][35e15eab81835bd28f13fabf8f2918bc] [0507b768564d58d088fb0bacfb8c4d42]

Type	Name	Papers
Journal	<i>Symmetry</i>	[f5d019cbcdce56b9a1f3e92f14ae5407][f19beaabe47855e396d12891544de9c7] [439eb98ec8385dd9adbb3b1f21a91e67]

FIGURE 6 Authors and journals appearing most in included papers

4. Discussion

The clearest substantive answer is that MCDM/AHP laptop-selection models usually combine **market-facing criteria** with **hardware criteria**. The market-facing side includes **price** and **brand image/reputation**; the hardware side includes **speed/processor**, **RAM**, **storage**, and **display/monitor** (Çakır & Pekkaya, 2020; Goswami & Behera, 2021). This pattern is consistent across laptop-specific and adjacent consumer-electronics studies, which strengthens confidence that these are the core criterion families rather than idiosyncratic one-off choices (Lam et al., 2023).

The evidence is weaker on a single best ranking. One direct laptop study elevates **brand image**, while adjacent studies often elevate **technical specification** or cost-related sub-criteria (Çakır & Pekkaya, 2020; Lam et al., 2023). That variation is expected because AHP weights reflect decision-maker preferences, and multiple reviews note that results depend on the weighting method, consistency of pairwise comparisons, and the chosen set of criteria (Youssef, 2020; Vinogradova et al., 2021; Erdebilli et al., 2023).

A second limitation is criteria selection itself. Several reviews show that MCDM studies often rely on literature review and expert opinion to define criteria, but many do not document that step rigorously (Shao et al., 2020; Liou et al., 2021; Theunissen et al., 2024). That matters here because “technical specification” can be modeled as one criterion in one paper and split into processor, RAM, storage, and battery in another.

A final methodological point is that classic AHP remains dominant for weighting, but many papers adopt hybrids or alternatives to reduce comparison burden or inconsistency (Pamucar et al., 2018; Youssef, 2020).

Claim	Evidence Strength	Reasoning	Papers
Price/cost is a recurring criterion in laptop and adjacent device-selection models.	 Strong	Directly named in laptop, smartphone, and broader MCDM selection studies.	(Çakır & Pekkaya, 2020; Goswami & Behera, 2021; Hazza et al., 2022)
Brand/reputation is commonly included, and can rank first in laptop models.	 Moderate	Strong direct laptop evidence, plus repeated use in adjacent electronics models.	(Çakır & Pekkaya, 2020; Lam et al., 2023; Goswami & Behera, 2021)
Technical specifications are usually represented either broadly or as sub-criteria like RAM, processor, and storage.	 Strong	Consistent across direct and adjacent product-selection models.	(Çakır & Pekkaya, 2020; Lam et al., 2023)
A universal ranking of laptop criteria is not stable across models .	 Moderate	Rankings differ by decision context and weighting choices.	(Çakır & Pekkaya, 2020; Lam et al., 2023; Vinogradova et al., 2021)
Formal criteria-selection procedures are often underdeveloped or underreported.	 Weak	Supported mainly by cross-domain methodological reviews rather than laptop-specific audits.	(Theunissen et al., 2024; Liou et al., 2021)

FIGURE 7 Key claims and supporting evidence in this corpus

5. Conclusion

The literature converges on a compact answer: MCDM and AHP models for laptop selection most commonly use **price/cost, brand image or reputation, technical performance, memory/storage, display/monitor**, and **other physical or peripheral features** (Çakır & Pekkaya, 2020). In practice, those broad criteria are often expanded into sub-criteria such as **RAM, processor, battery power, and internal storage**, especially in adjacent consumer-electronics models (Lam et al., 2023; Goswami & Behera, 2021).

Research Gaps

The main gap is not the absence of candidate criteria, but the limited standardization of **how criteria are selected, grouped, and validated** across studies. Laptop-specific papers are sparse, and much of the transferable evidence comes from smartphones, suppliers, or cloud-service selection rather than notebooks themselves (Theunissen et al., 2024; Balasbaneh et al., 2025).

Topic/Outcome	Study-specific laptop evidence	Adjacent electronics evidence	Formal criteria selection	Weighting sensitivity
Price and affordability	2	4	1	3
Brand and reputation	2	4	1	2
Technical specifications	2	5	1	3
Display and physical design	1	3	GAP	1
Sustainability or lifecycle features	GAP	1	1	1

Open Research Questions

Future work should test whether laptop-specific criteria differ meaningfully across user segments and whether data-driven feature extraction changes criterion sets compared with expert-defined hierarchies.

Question	Why
Do office, student, and gaming laptop buyers require different AHP criterion hierarchies?	Segment-specific priorities likely differ, but current laptop-specific MCDM evidence is too sparse to define stable hierarchies.
Does data-driven feature extraction outperform expert-only criteria selection in laptop MCDM models?	Reviews suggest criteria are often expert-defined and underreported, leaving uncertainty about omitted or over-aggregated laptop features.
Should sustainability and repairability be standard criteria in future laptop-selection MCDM frameworks?	Sustainability criteria are common in other MCDM domains but rarely appear in laptop-selection models despite growing relevance.

Overall, the most common MCDM/AHP criteria for laptop selection are **price, brand, technical specs, storage/memory, display, and related physical features**, with exact rankings varying by model and decision context.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Altubaishe, B., & Desai, S. (2023). Multicriteria Decision Making in Supply Chain Management Using FMEA and Hybrid AHP-PROMETHEE Algorithms. *Sensors (Basel, Switzerland)*, 23. <https://doi.org/10.3390/s23084041>

- Balasbaneh, A. T., Aldrovandi, S., & Sher, W. (2025). A Systematic Review of Implementing Multi-Criteria Decision-Making (MCDM) Approaches for the Circular Economy and Cost Assessment. *Sustainability*. <https://doi.org/10.3390/su17115007>
- Çakır, F. S., & Pekkaya, M. (2020). Determination of Interaction Between Criteria and the Criteria Priorities in Laptop Selection Problem. *International Journal of Fuzzy Systems*, 22, 1177 - 1190. <https://doi.org/10.1007/s40815-020-00857-2>
- Dahooie, J. H., Raafat, R., Qorbani, A., & Daim, T. (2021). An intuitionistic fuzzy data-driven product ranking model using sentiment analysis and multi-criteria decision-making. *Technological Forecasting and Social Change*, 173, 121158. <https://doi.org/10.1016/j.techfore.2021.121158>
- Erdebilli, B., Yilmaz, I., Aksoy, T., Hacıoglu, U., Yüksel, S., & Dinçer, H. (2023). An Interval-Valued Pythagorean Fuzzy AHP and COPRAS Hybrid Methods for the Supplier Selection Problem. *International Journal of Computational Intelligence Systems*, 16. <https://doi.org/10.1007/s44196-023-00297-4>
- Goswami, S., & Behera, D. (2021). An Analysis for Selecting Best Smartphone Model by AHP-TOPSIS Decision-Making Methodology. *Int. J. Serv. Sci. Manag. Eng. Technol.*, 12, 116-137. <https://doi.org/10.4018/ijssmet.2021050107>
- Gyani, J., Ahmed, A., & Haq, M. A. (2022). MCDM and various prioritization methods in AHP for CSS: A comprehensive review. *IEEE Access*, PP, 1-1. <https://doi.org/10.1109/access.2022.3161742>
- Hazza, M. A. A., Abdelwahed, A., Ali, M. Y., & Sidek, A. A. (2022). An Integrated Approach for Supplier Evaluation and Selection using the Delphi Method and Analytic Hierarchy Process (AHP): A New Framework. *International Journal of Technology*. <https://doi.org/10.14716/ijtech.v13i1.4700>
- Jawad, M., Naz, M., & Muqaddus, H. (2023). A multi-criteria decision-making approach for portfolio selection by using an automatic spherical fuzzy AHP algorithm. *Journal of the Operational Research Society*, 75, 85 - 98. <https://doi.org/10.1080/01605682.2023.2174905>
- Lam, W. S., Lam, W. H., Liew, K. F., Bakar, M. A., & Lai, C. P. (2023). Evaluation and Selection of Mobile Phones using Integrated AHP-TOPSIS Model. *Journal of Advanced Research in Applied Sciences and Engineering Technology*. <https://doi.org/10.37934/araset.33.2.2539>
- Liou, J., Chang, M., Lo, H.-W., & Hsu, M. (2021). Application of an MCDM model with data mining techniques for green supplier evaluation and selection. *Appl. Soft Comput.*, 109, 107534. <https://doi.org/10.1016/j.asoc.2021.107534>
- Pamucar, D., Stević, Ž., & Sremac, S. (2018). A New Model for Determining Weight Coefficients of Criteria in MCDM Models: Full Consistency Method (FUCOM). *Symmetry*, 10, 393. <https://doi.org/10.3390/sym10090393>
- Pramanik, P. K., Biswas, S., Pal, S., Marinković, D., & Choudhury, P. (2021). A Comparative Analysis of Multi-Criteria Decision-Making Methods for Resource Selection in Mobile Crowd Computing. *Symmetry*, 13, 1713. <https://doi.org/10.3390/sym13091713>
- Rahardjo, B., Wang, F., Lo, S.-C., & Chou, J.-H. (2023). A Hybrid Multi-Criteria Decision-Making Model Combining DANP with VIKOR for Sustainable Supplier Selection in Electronics Industry. *Sustainability*. <https://doi.org/10.3390/su15054588>
- Rasmussen, A., Sabic, H., Saha, S., & Nielsen, I. (2022). Supplier selection for aerospace & defense industry through MCDM methods. *Cleaner Engineering and Technology*. <https://doi.org/10.1016/j.clet.2022.100590>
- Shao, M., Han, Z., Sun, J., Xiao, C., Zhang, S.-L., & Zhao, Y.-X. (2020). A review of multi-criteria decision making applications for renewable energy site selection. *Renewable Energy*. <https://doi.org/10.1016/j.renene.2020.04.137>
- Theunissen, F., Bezuidenhout, C., & Alam, S. (2024). Exploring the shortcomings in formal criteria selection for multicriteria decision making based inventory classification models: a systematic review and future directions. *International Journal of Production Research*, 62, 7279 - 7299. <https://doi.org/10.1080/00207543.2024.2320680>
- Villalba, P., Sánchez-Garrido, A. J., & Yepes, V. (2024). A REVIEW OF MULTI-CRITERIA DECISION-MAKING METHODS FOR BUILDING ASSESSMENT, SELECTION, AND RETROFIT. *JOURNAL OF CIVIL ENGINEERING AND MANAGEMENT*. <https://doi.org/10.3846/jcem.2024.21621>
- Vinogradova, I., Podvezko, V., & Zavadskas, E. (2021). Comparative Assessment of the Stability of AHP and FAHP Methods. *Symmetry*, 13, 479. <https://doi.org/10.3390/sym13030479>
- Yannis, G., Kopsacheili, A., Dragomanovits, A., & Petraki, V. (2020). State-of-the-art review on multi-criteria decision-making in the transport sector. *Journal of Traffic and Transportation Engineering (English Edition)*. <https://doi.org/10.1016/j.jtte.2020.05.005>
- Youssef, A. E. (2020). An Integrated MCDM Approach for Cloud Service Selection Based on TOPSIS and BWM. *IEEE Access*, 8, 71851-71865. <https://doi.org/10.1109/access.2020.2987111>

Zakeri, S., Chatterjee, P., Konstantas, D., & Ecer, F. (2023). A decision analysis model for material selection using simple ranking process. *Scientific Reports*, 13. <https://doi.org/10.1038/s41598-023-35405-z>